



PURPLE INDUSTRIES · KUBERNETES LEARNER CONSOLE

Kubernetes

A hands-on CKA practice simulator

Inject faults, repair workloads, and get graded against real Kubernetes API behavior — no cloud account required.



purpleindustries.us/kubernetes



github.com/BarnsL/Kubernetes-Simulator



What is Kubernetes?

Kubernetes (K8s) is an open-source platform that automates the deployment, scaling, and operation of containerized applications across a cluster of machines.



You declare the desired state — Kubernetes makes it real

Tell the cluster "run 3 replicas of this image" and the control plane continuously reconciles reality toward that goal — restarting, rescheduling, and replacing as needed. A CNCF graduated project and the industry-standard container orchestrator.



Self-healing

Restarts, reschedules, and replaces unhealthy Pods automatically



Declarative

Desired state in YAML; controllers reconcile continuously



Elastic scaling

Scale workloads up or down on demand or automatically



Service discovery

Built-in DNS, networking, and load balancing for Pods

The core building blocks



Pod

Smallest deployable unit — one or more containers



Deployment

Declarative rollouts over self-healing
ReplicaSets



Service

Stable network identity for a set of Pods



ConfigMap & Secret

Externalized configuration and sensitive data



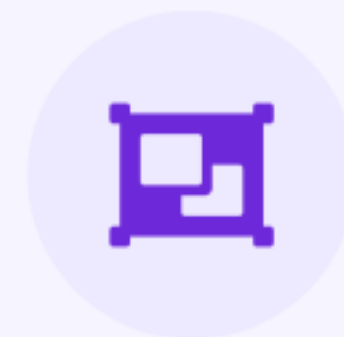
PV & PVC

Persistent storage requests bound to volumes



RBAC

Roles and bindings that authorize every action



Namespace

Scoping and isolation boundaries in a cluster



Node

The worker machines that actually run Pods

Meet the CKA Practice Simulator

A Go CLI and web GUI for practicing Certified Kubernetes Administrator exam scenarios against a real or simulated cluster.



32 authentic mission labs

Pods, networking, storage, RBAC, scheduling & more



Inject → repair → grade loop

The same shape as real CKA exam tasks



Offline fake-client mode

Runs with no cluster and no cloud account

```
cka-sim - self-test

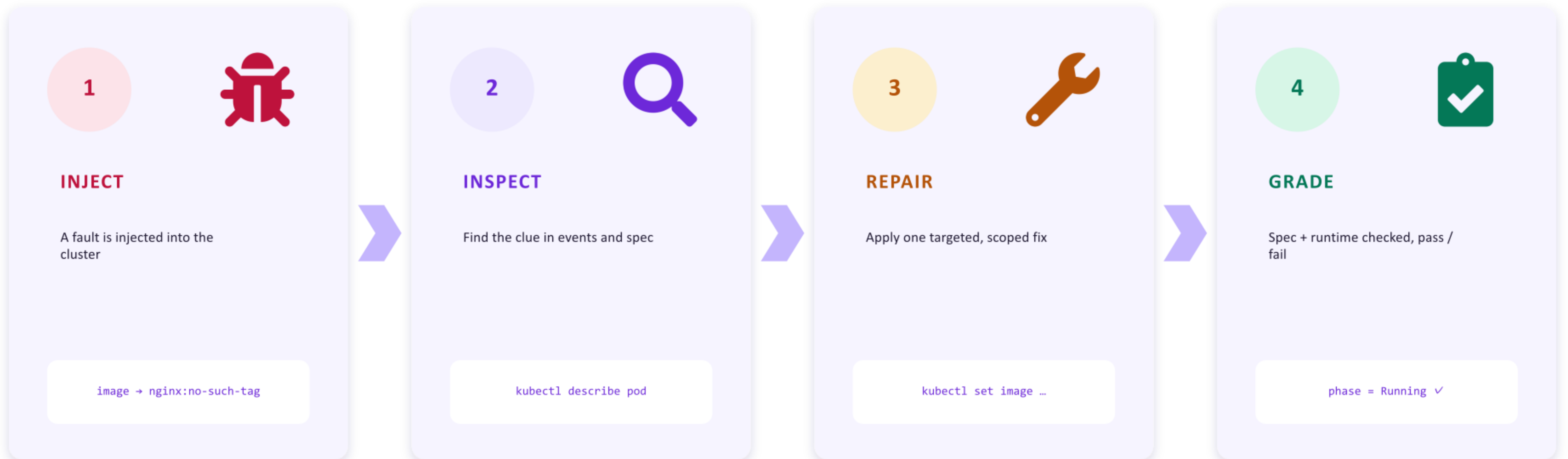
$ go build -o cka-sim ./cmd/cka-sim
$ ./cka-sim self-test

▶ scenario injected: demo-pod
image set to nginx:no-such-tag
status: Pending

✓ repair applied
image → nginx:1.25
✓ pod reached Running

RESULT: PASS
```


How every mission works



You don't just read about Kubernetes — you break it, diagnose it, and fix it like an operator.

Architecture



repo layout

```
cmd/cka-sim/ CLI entrypoint
internal/
grader/ pod state · image · readiness
injector/ broken images · misconfigs
scenarios/ scenario definitions
selftest/ end-to-end fake-client flow
web/ React + TypeScript GUI (Vite)

go test ./... all green
```



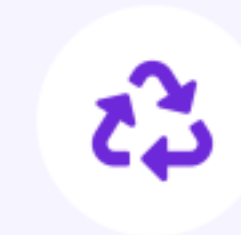
Separation of concerns

Scenario definitions live apart from grading logic



API is the source of truth

Validation reads real Kubernetes object state



Disposable-cluster first

Designed to run against throwaway local clusters



Offline by default

client-go/fake powers practice with no cluster

Practice offline, or grade a real cluster



Fake-client mode

OFFLINE

Runs entirely offline using client-go/fake
No cluster, no kubeconfig, no cloud account
Deterministic, repeatable scenarios
Study anywhere — on a plane, offline

VS



Live mode

REAL CLUSTER

Grades against a real kubeconfig cluster
Use kind, minikube, or a full cluster
Builds real kubectl muscle memory
Closest to actual exam conditions

Coverage across the CKA curriculum

32 labs



Workloads & controllers

Pod image · Deployment rollout · readiness probe · CrashLoopBackOff · Job · CronJob · DaemonSet · StatefulSet



Config & storage

ConfigMap key · Secret env · resource requests · PV/PVC binding · finalizer troubleshooting



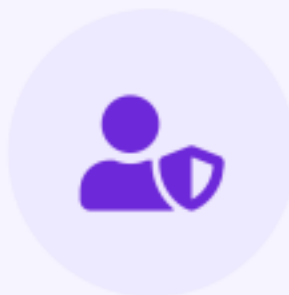
Networking & services

Service selector · Ingress backend · NetworkPolicy · EndpointSlice · DNS policy · cross-namespace lookup



Scheduling

Node scheduling clinic · taint & toleration match · label selector repair



Security & RBAC

RBAC binding · ServiceAccount binding · token mount control · Pod Security Baseline



Cluster setup & tooling

kubectl verify · kind bootstrap · minikube · Hello Minikube · kubeadm path · namespace cleanup

Pod image repair

BEGINNER

WORKLOADS

Identify a bad image value and recover a Pod to Running.

A Pod is stuck because its container image tag is invalid. The fix maps directly to one failing field — but a pass needs both the right image and a healthy runtime state.



BROKEN STATE
demo-pod 0/1 Pending · nginx:no-such-tag



inspect → repair → verify

```
$ kubectl describe pod demo-pod
Image: nginx:no-such-tag
Events: Failed to pull image

$ kubectl set image pod/demo-pod \
app=nginx:1.25
pod/demo-pod image updated

$ kubectl wait --for=condition=Ready \
pod/demo-pod
pod/demo-pod condition met ✓
```

PASS CRITERIA

spec.image = nginx:1.25

status.phase = Running

The web learning console

Learn Kubernetes concepts while you practice — terminal-first, with explanations at every step.



Simulated kubectl terminal

Run modeled commands safely against simulated state



Token-by-token breakdown

Every part of a command explained as you type it



Step-by-step tutorials

A guided inspect → repair → verify learning path



Mini cluster dashboard

Overview, workloads, services, config & storage, cluster



Glossary & field guide

Understand exactly what the grader is checking

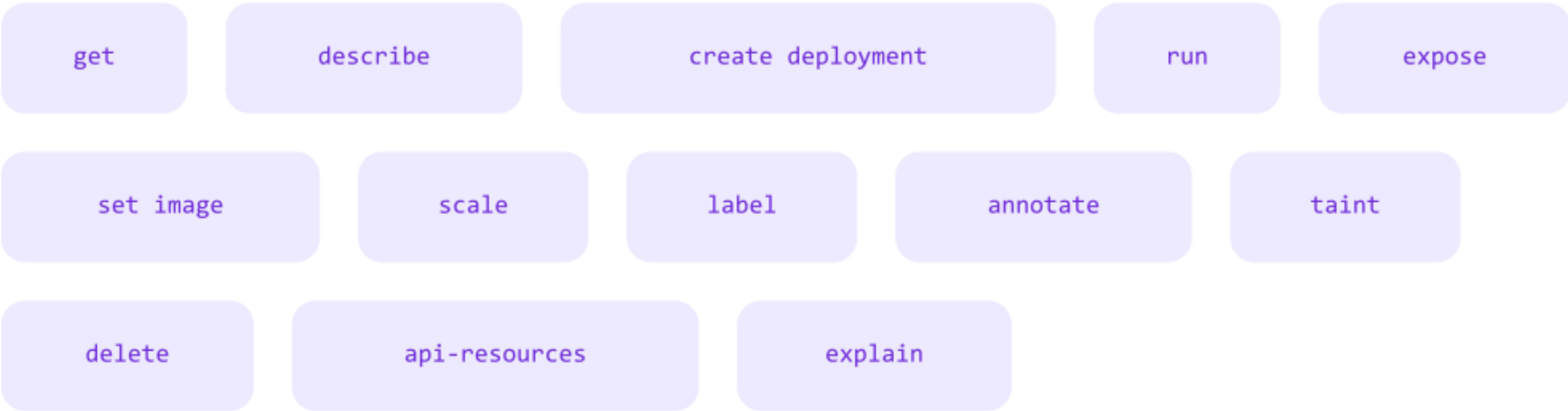


Free-play sandbox

A mutable cluster for common kubectl verbs

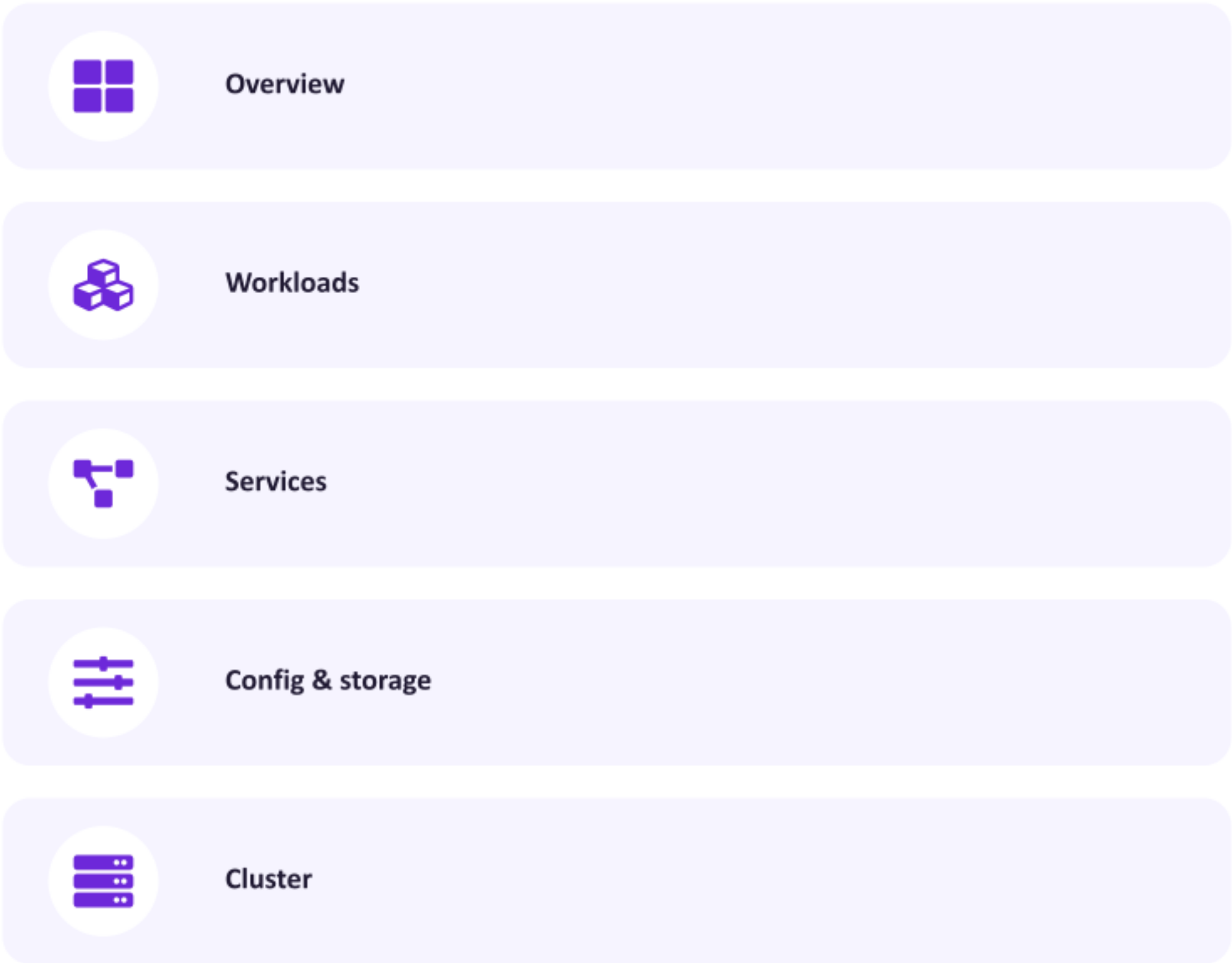
A sandbox with real cluster state

Modeled kubectl verbs



The simulated cluster has state. Later reads reflect earlier mutations, so create → scale → get behaves like a real session — not canned output.

Dashboard information architecture



Your latest command maps to the section that matters next.

Get started in minutes



Command-line (Go)

1

Build the CLI

```
go build -o cka-sim ./cmd/cka-sim
```

2

Run the self-test

```
./cka-sim self-test
```

3

Inject, then grade

```
./cka-sim inject ... && ./cka-sim grade ...
```



Web console

1

Enter the web app

```
cd web
```

2

Install dependencies

```
npm install
```

3

Start the dev server

```
npm run dev
```

Go 1.21+

Node.js 18+

kubectl (live mode)

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Break it. Fix it. Pass.

Hands-on Kubernetes practice that mirrors the real CKA exam — inject a fault, repair the workload, and let the grader confirm you got it right.



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